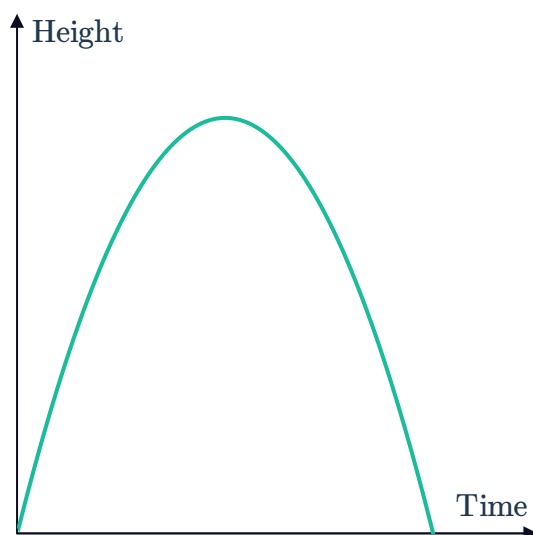


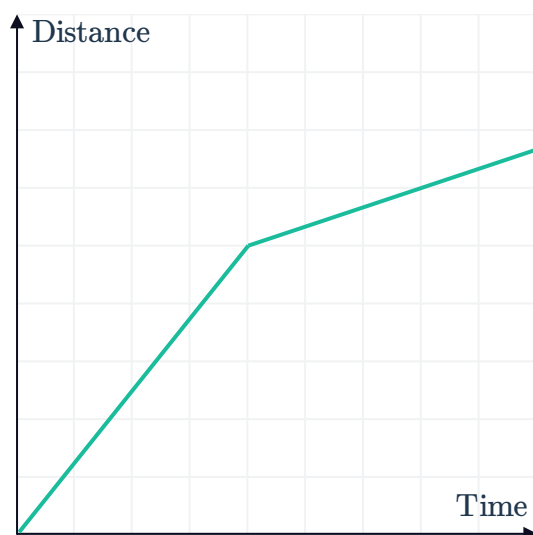
Travel graphs



1



2 Example answer:



3 Example answer:



4 Distance-Time graphs for objects moving with constant acceleration



5

- a 150 km
- c 50 km/h
- e 250 km
- b Ben decreased his speed at 11 am.
- d 0 km
- f 500 km

6

- a 120 km
- b 7 hours
- c 1:30 pm - 2 pm
- d 120 km/h
- e 100 km
- f 180 km

7

- a 5 km
- b 30 km
- c 1 hour
- d 10 km
- e First two hours.

8

- a 9 hours
- b 110 km
- c 12.22 km/h

9

- a Towards
- b 1 minute
- c 2 minutes
- d 4 km
- e 7 minutes
- f 9 km

10

- a 3 km/min
- b 1.2 km/min
- c 1 km/min

11

- a 12:00 pm
- b 1:00 pm
- c 80 km
- d 260 km
- e 9:00 am til 10:00 am

12

- a 50 km/h
- b Between 2 and 3 pm
- c Stopping for rest

13

- a 2 km
- b 1:00 pm
- c 4 km

14

- a 2100 m
- b 35 minutes
- c 1500 m
- d 20 minutes

15



- a 60 minutes
- b Between 10 and 20 minutes, and between 30 and 40 minutes.

16

- a Example answer:

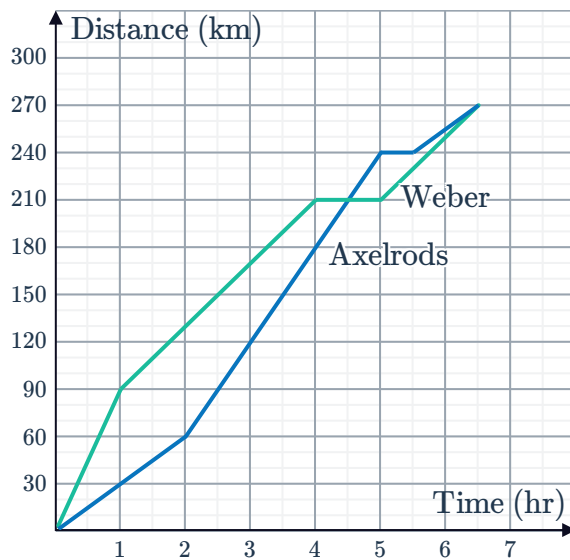
Xanthe is playing with a remote-controlled toy car. She holds down the acceleration button and the toy car goes forwards, continuing to increase in speed until it runs into a wall. It takes Xanthe a few moments to find the switch for 'reverse', during which time the toy car remains against the wall. After flipping the switch, the toy car accelerates back in the direction it came from, until it reaches Xanthe.

- b Example answer:

Sophia and a group of friends decide to go on a bushwalk. They walk for a while and then stop in a clearing to have lunch. They continue to walk for some time before they come to the edge of a cliff. At this point, they take a sharp turn and begin to walk back in the direction they came from.

17

- a 9:30 am
- b 210 km
- c 7:00 am
- d 4:00 am and 7:00 am
- e Example answer:



Graph comparisons

18 Ray, by 20 km/h

19 Oliver, 10 km/h

20 a 2000 m

b 100 m



c 200 m

d 12 minutes

21

a 250 m

b 150 m

c 50 m/min

d 20 m/min

e 60 minutes

22

a 1.5 hours

b 6.67 km/h

c 20 km/h

d 11:15 am

e 15 km

23

a 200 km

b 11:00 am

c 150 km

d 4:00 pm

e 1 hour